



PHYS 551 – Advanced Monte Carlo Methods

Spring 2025/26

TERM PROJECT

**Rare-Event Estimation of Cascading Blackouts in the
Istanbul Power Grid via Importance Sampling**

Submitted by: Ömer Faruk Avcı

Student ID: S034113

Course Instructor: Prof. Taylan Akdoğan

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1 Introduction

A large electrical grid does not fail one component at a time. When a single district is pushed past the limit of its substations, its excess load does not simply vanish; it is rerouted onto the neighbouring districts. If those neighbours are themselves close to their limit, the overload spreads, and a local fault grows into a city-wide blackout. This chain reaction is called a *cascading failure*. Because the number of possible failure combinations in a grid of tens of districts is enormous, and because the truly catastrophic events are very rare, deterministic worst-case analysis is not practical. This term project therefore studies the problem with Monte Carlo simulation.

The grid studied here is the mainland of Istanbul, modelled as 38 districts on the European and Asian sides. Each district has a population, a base electricity load, and a safety margin that is calibrated from real overload records published by EPIAŞ. The central difficulty is that a blackout affecting millions of people is a *rare event*: under normal operation it almost never happens, so a naive (crude) Monte Carlo simulation would have to run for an impractically long time before it ever observed one. The project answers two questions:

1. What is the probability that a cascading failure leaves at least X people without power on a given day, including the rare tail that naive Monte Carlo cannot reach? I answer this with Importance Sampling.
2. Given the resulting risk map, *where* should new relief substations be built to reduce the expected number of people affected the most?

The remainder of this report is organised as follows. Section 2 describes the data and the model: the districts, how loads and safety margins are obtained, and the cascading-failure mechanism. Section 3 presents the naive Monte Carlo baseline and shows where it breaks down. Section 4 introduces the Importance Sampling estimator and the systemic-surge bias that makes the rare tail tractable. Section 5 compares the two methods directly through the exceedance curve, the variance-reduction table, and a convergence study. Section 6 applies the model to the substation-placement question. Section 7 concludes.

Declaration of AI Usage

In compliance with the course’s principles for the responsible use of AI, I disclose my use of AI tools in this project. The Python code was written by an AI assistant following my prompts and direction. The core of the work, however, is my own: the design of the algorithm and how the model works – the cascading-failure mechanism, the systemic-surge demand model, and the Importance Sampling scheme – together with the calibration choices and the sourcing of the real data (the EPIAŞ electricity-consumption and overload “Aşırı Yük” records that set the district loads and safety margins) were all done by me. I directed the AI at each step and reviewed and verified every result against the simulation before including it.

2 System Model and Data

2.1 Districts, Population, and Load

The grid is represented as a graph of 38 mainland districts (the islands are excluded), with a total population of about 15 736 564. Two districts are connected by an edge if they share a physical border, giving 83 edges; the adjacency was extracted from district geometry and adjusted for the Bosphorus bridge crossings. Figure 1 shows the resulting network.

Each district is assigned a base electricity load μ_i . The total monthly consumption of the European and Asian distribution regions (from EPIAŞ consumption data) is distributed among the districts of each

side in proportion to their population:

$$\mu_i = C_{\text{side}} \cdot \frac{\text{Pop}_i}{\sum_{j \in \text{side}} \text{Pop}_j}, \quad (1)$$

where C_{side} is the average monthly consumption of the side that district i belongs to. This makes the load distribution reflect the real demand split between the two sides of the city. Because both capacity and demand are proportional to μ_i (Section 2.3), the model is scale-free: only dimensionless ratios enter the results, so the absolute size of μ_i and its time unit are immaterial.

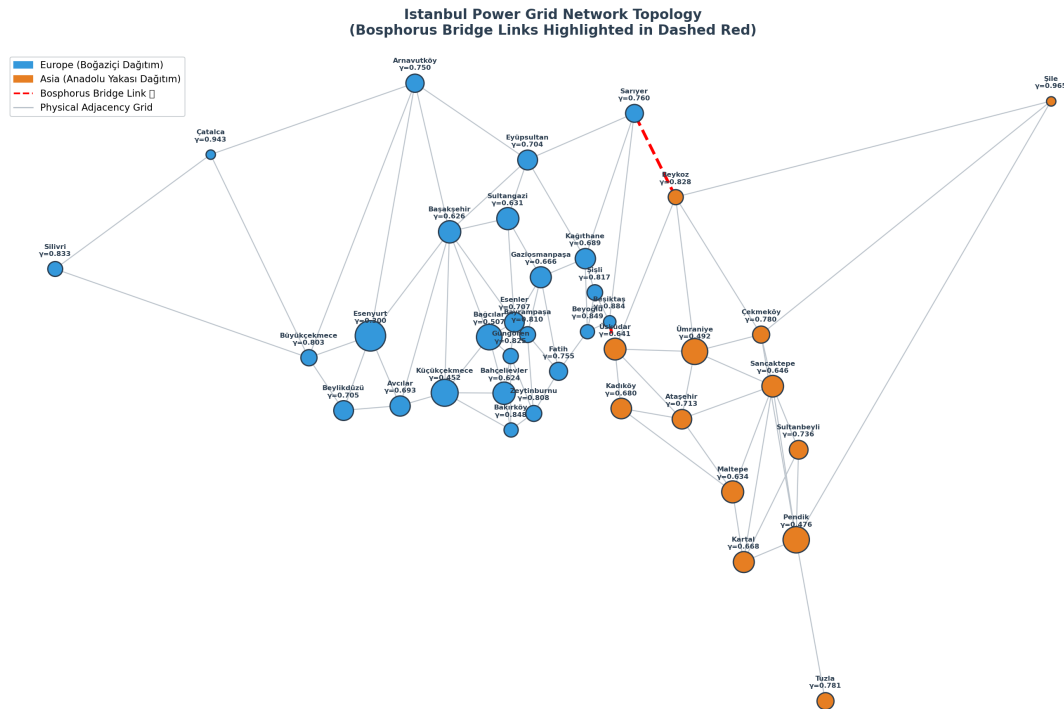


Figure 1: The Istanbul mainland grid as a graph of 38 districts. Nodes are placed at district centroids; edges connect bordering districts.

2.2 Safety Margins from Real Overload Data

Each district is given a dimensionless safety margin α_i , defined so that its capacity is

$$\text{capacity}_i = \mu_i (1 + \alpha_i). \quad (2)$$

A district with a large α_i can absorb a big demand spike before tripping; a district with a small α_i is fragile. Rather than assign these margins arbitrarily, they are calibrated from one year of EPIAŞ outage records by counting, for each district, how many outages were caused by overload (“Aşırı Yük”). The margins are then assigned *per side* (because the two sides are operated by different distribution companies) by linearly mapping the overload count to the range $[\alpha_{\min}, \alpha_{\max}] = [0.08, 0.25]$:

$$\alpha_i = \alpha_{\max} - \frac{n_i}{\max_{j \in \text{side}} n_j} (\alpha_{\max} - \alpha_{\min}), \quad (3)$$

where n_i is the overload count of district i . The districts with the most recorded overloads are therefore given the smallest margins. For example, Maltepe (465 overloads, the worst on the Asian side) and Fatih and Beyoğlu (108 each, the worst on the European side) receive $\alpha = 0.08$, while a quiet district such as Çatalca (2 overloads) is close to $\alpha = 0.25$. This is what makes the model data-driven: the historically weakest districts are the structurally weakest in the simulation.

2.3 The Cascading-Failure Mechanism

Daily demand. Electricity demand fluctuates from day to day. The demand of district i on a given day is modelled as

$$L_i = \mu_i (S + \varepsilon_i), \quad (4)$$

where $S \sim \mathcal{N}(1, \sigma_{\text{sys}}^2)$ is a city-wide demand multiplier shared by every district, and $\varepsilon_i \sim \mathcal{N}(0, \sigma_{\text{idio}}^2)$ is a small district-specific fluctuation. The systemic term S represents a common cause, such as a heatwave that turns on every air conditioner in the city at once; it is the reason failures are correlated, and a rare large value of S is what drives a city-wide cascade. I use $\sigma_{\text{sys}} = 0.045$ and $\sigma_{\text{idio}} = 0.02$.

Tripping and redistribution. A district trips when its demand exceeds its capacity, i.e. when $L_i > \text{capacity}_i$. This breaker is a hard threshold, so once the demand vector is drawn the whole cascade is deterministic; all randomness lives in S and the ε_i . When district i trips, a fraction of its excess load is pushed onto its still-active neighbours:

$$\text{load to move} = \gamma_i (L_i - \text{capacity}_i), \quad (5)$$

where $\gamma_i \in (0, 1]$ is a redistribution factor. The load is split among the live neighbours in proportion to each neighbour's remaining headroom, so load preferentially flows to the districts most able to absorb it. A neighbour that is itself pushed over its capacity then trips in turn, and the process repeats until no further districts fail. The redistribution factor is kept deliberately moderate ($\gamma_i \leq 0.35$) so that a single trip does not automatically collapse the whole city; a truly large blackout requires a large systemic surge S , which is rare. The quantity of interest at the end of each simulated day is the total population without power, i.e. the sum of the populations of all tripped districts.

2.4 Summary of Notation

Table 1 collects the symbols used throughout the report.

Table 1: Notation.	
Symbol	Meaning
Pop_i	population of district i
μ_i	base (mean) electricity load of district i
α_i	safety margin; capacity is $\mu_i(1 + \alpha_i)$
γ_i	fraction of excess load district i sheds to neighbours
S	city-wide demand surge, $\mathcal{N}(1, \sigma_{\text{sys}}^2)$
ε_i	district demand noise, $\mathcal{N}(0, \sigma_{\text{idio}}^2)$
$\sigma_{\text{sys}}, \sigma_{\text{idio}}$	surge / noise std. (0.045, 0.02)
$L_i = \mu_i(S + \varepsilon_i)$	sampled per-step demand of district i
θ	importance-sampling tilt of S (in std. units)
W	importance weight f/g

3 Naive Monte Carlo Baseline

The naive (crude) estimator samples the model exactly as described above: for each of N simulated days, a systemic surge S and the district fluctuations ε_i are drawn from their true distributions, the cascade is run, and the affected population is recorded. The probability that at least X people lose power is estimated by the fraction of days on which this happened,

$$\hat{P}(\text{affected} \geq X) = \frac{1}{N} \sum_{k=1}^N \mathbf{1}\{\text{affected}_k \geq X\}, \quad (6)$$

with the usual binomial standard error $\sqrt{\hat{P}(1 - \hat{P})/N}$.

Table 2 shows the result of a run with $N = 100000$ days. The estimator is accurate for the common, small events: outages affecting more than 100000 people occur on about 9% of days, and the curve is well-resolved down to the 2000000-person level (277 observed events). But the deep tail reveals the central problem: in 100000 simulated days, only two events affecting more than 8000000 people were observed. Two events give a hopelessly noisy estimate (a relative error near 70%), and a slightly rarer event would not have been seen at all. The naive method cannot reliably reach the events that matter most.

Table 2: Naive Monte Carlo exceedance probabilities, $N = 100000$ days.

Threshold X (people)	Observed events	$\hat{P}(\text{affected} \geq X)$
100000	9343	9.34×10^{-2}
1000000	2282	2.28×10^{-2}
2000000	277	2.77×10^{-3}
4000000	15	1.50×10^{-4}
6000000	3	3.00×10^{-5}
8000000	2	2.00×10^{-5} (only 2 events)

4 Importance Sampling

4.1 Idea and Estimator

Importance Sampling removes the blindness of the naive method by drawing samples from a different, biased distribution g that makes the rare event common, and then correcting for the bias with a weight. If the true distribution is f and I instead sample from a proposal g , the unbiased estimator of any probability is

$$\hat{P} = \frac{1}{N} \sum_{k=1}^N \mathbf{1}\{\text{event}_k\} W_k, \quad W_k = \frac{f(x_k)}{g(x_k)}. \quad (7)$$

The weight W_k compensates for the fact that the biased distribution visits the rare region more often than reality would.

4.2 Biasing the Systemic Surge

The key modelling insight is that, in this grid, almost all of the tail risk is driven by the single systemic variable S : a city-wide blackout happens essentially only when S is large. I therefore bias *only* S , shifting its mean upward by θ standard deviations,

$$g: \quad S \sim \mathcal{N}(1 + \theta \sigma_{\text{sys}}, \sigma_{\text{sys}}^2), \quad (8)$$

while the district fluctuations ε_i are left untouched, and the cascade itself is deterministic given the demand, so neither needs reweighting. Because the bias acts on a single dimension, the likelihood ratio is just a ratio of two Gaussian densities and collapses to a closed form. Writing $z_S = (S - 1)/\sigma_{\text{sys}}$,

$$W = \frac{f(S)}{g(S)} = \frac{\exp(-\frac{1}{2} z_S^2)}{\exp(-\frac{1}{2} (z_S - \theta)^2)} = \exp(-\theta z_S + \frac{1}{2} \theta^2), \quad \ln W = -\theta z_S + \frac{1}{2} \theta^2. \quad (9)$$

The weight depends on the draw only through the single number z_S , so it stays well-behaved. This is far more stable than biasing all 38 districts independently, where the weight would be a product of 38 such factors and would degenerate.

Choosing the tilt. For estimating a Gaussian-tail probability $P(S > c)$, the variance-minimising exponential tilt shifts the proposal mean onto the threshold itself, $\theta^* \approx (c - 1)/\sigma_{\text{sys}}$ – the z -score of the event. A more severe blackout corresponds to a larger surge c , hence a larger optimal θ ; this is why no single tilt is best for the whole curve and a ladder of tilts is needed.

A single value of θ samples only one part of the tail well, so I run a *ladder* of tilts $\theta \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ with 80000 samples each and combine them with *multiple importance sampling* (the balance heuristic): every draw, whichever tilt produced it, is reweighted by

$$w(S) = \frac{f(S)}{\bar{g}(S)}, \quad \bar{g}(S) = \frac{1}{K} \sum_{l=1}^K g_l(S), \quad (10)$$

the ratio of the true density to the equal mixture of the $K = 5$ proposals. All $KN = 400000$ draws are pooled into one unbiased estimator $\hat{P}(\geq t) = \frac{1}{KN} \sum \mathbf{1}\{\cdot\} w$. This avoids the bias of selecting a “best” tilt per threshold and yields one honest effective sample size for the whole pool. The rarest bucket (≥ 8000000 people), which the naive method barely saw, is estimated as

$$\hat{P}(\text{affected} \geq 8000000) = (5.22 \pm 0.07) \times 10^{-6}, \quad (11)$$

a one-in-191000-day event, with a relative error near 1%. The pooled effective sample size is modest (about 0.4% of the draws), but that global figure is not the right yardstick here: because the high tilts place their samples exactly on the rare region, the *per-threshold* relative error is what matters, and it stays around 1% across the tail (Table 3).

4.3 A Closed-Form Check

Before trusting the simulated probabilities, it is worth checking the model against a hand calculation. Ignoring the cascade, the most fragile districts (Maltepe, Beyoğlu, Fatih, with $\alpha = 0.08$) fail exactly when their own demand exceeds capacity, i.e. when $S + \varepsilon_i > 1 + \alpha$. Since $S + \varepsilon_i \sim \mathcal{N}(1, \sigma_{\text{tot}}^2)$ with $\sigma_{\text{tot}} = \sqrt{\sigma_{\text{sys}}^2 + \sigma_{\text{idio}}^2} = \sqrt{0.045^2 + 0.02^2} \approx 0.049$, the direct-overload probability is

$$P_{\text{direct}} = \Phi\left(-\frac{\alpha}{\sigma_{\text{tot}}}\right) = \Phi(-1.63) \approx 5.2\%. \quad (12)$$

The simulation gives 5.2% for these districts – essentially identical, which validates it against an independent calculation. The match also shows that the cascade adds almost nothing to a single fragile district’s *own* probability: when such a district trips, its excess is absorbed by well-margined neighbours rather than tipping them. The cascade’s real effect is instead to make failures *correlated*, and it is that correlation – many districts going down on the same large- S day – that populates the deep tail of the population curve.

5 Comparison of the Two Methods

5.1 The Exceedance Curve

Figure 2 overlays the two estimators. On the common events (left side of the curve) the naive and Importance Sampling estimates lie exactly on top of each other – this agreement is the essential check that the Importance Sampling estimator is unbiased, since a wrong implementation would disagree here. As events become rarer the naive error bars explode: by the 8000000-person level the naive estimate rests on just two observed events and is effectively useless, whereas Importance Sampling stays smooth and tightly bounded down to $\sim 5 \times 10^{-6}$.

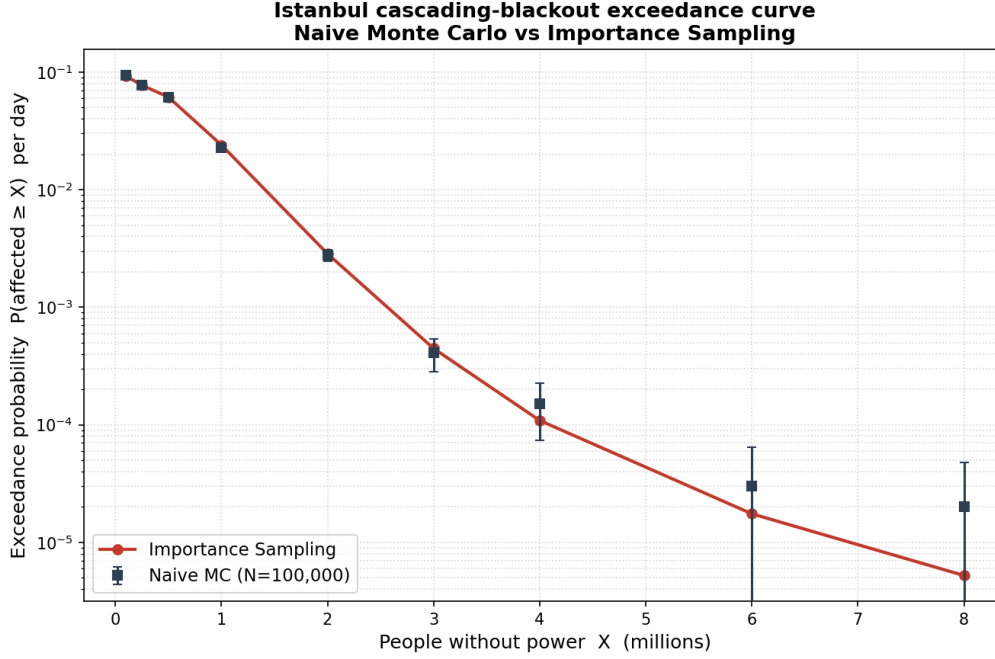


Figure 2: Exceedance probability $P(\text{affected} \geq X)$ per day. The two methods agree on common events; in the rare tail only Importance Sampling stays precise, while the naive estimate rests on a handful of events with very large error bars.

5.2 Variance Reduction

Table 3 quantifies the gain. The variance-reduction factor is the ratio of the per-sample variances of the two estimators, evaluated at the true probability. For common events the gain is negligible (the two estimators are essentially equivalent there), but it grows steeply as the event becomes rarer, reaching roughly $10000\times$ for the rarest bucket. The naive figure in the last row is itself unreliable: in 100000 days the naive run happened to log only two ≥ 8000000 events, giving $\hat{P} \approx 2 \times 10^{-5}$ with a 95% interval that runs from near zero to about four times the Importance Sampling value – it cannot tell 5×10^{-6} from 5×10^{-5} . The final column states the gain in the most intuitive way: to expect even a *single* ≥ 8000000 event, the naive method needs on the order of 190000 simulated days, whereas Importance Sampling resolves its probability to about 1% relative error from a pooled 400000 tilted draws.

Table 3: Per-threshold comparison of naive Monte Carlo and Importance Sampling, and the resulting variance-reduction factor.

Threshold X	Naive $\hat{P}$	IS $\hat{P}$	Var. reduction	Naive days / event
100000	9.34×10^{-2}	9.23×10^{-2}	$1\times$	11
1000000	2.28×10^{-2}	2.39×10^{-2}	$6\times$	42
2000000	2.77×10^{-3}	2.85×10^{-3}	$42\times$	351
4000000	1.50×10^{-4}	1.09×10^{-4}	$644\times$	9195
6000000	3.00×10^{-5}	1.75×10^{-5}	$3851\times$	57183
8000000	2.00×10^{-5}	5.22×10^{-6}	$10583\times$	191450

5.3 Convergence

Figure 3 tracks the running estimate of the rarest bucket against the number of samples. The Importance Sampling estimate settles smoothly onto its value of about 5.2×10^{-6} with a steadily shrinking

confidence band. The naive estimate, by contrast, sits at zero for the first $\sim 39\,000$ days, then jumps discontinuously each time it happens to record one of its two events and drifts at roughly 2×10^{-5} – about four times too high – with an error band that still spans zero at $N = 100\,000$. The naive curve is therefore not converging to anything useful; it is just sampling noise. This is a direct picture of why the rare-event problem requires variance reduction rather than simply more samples.

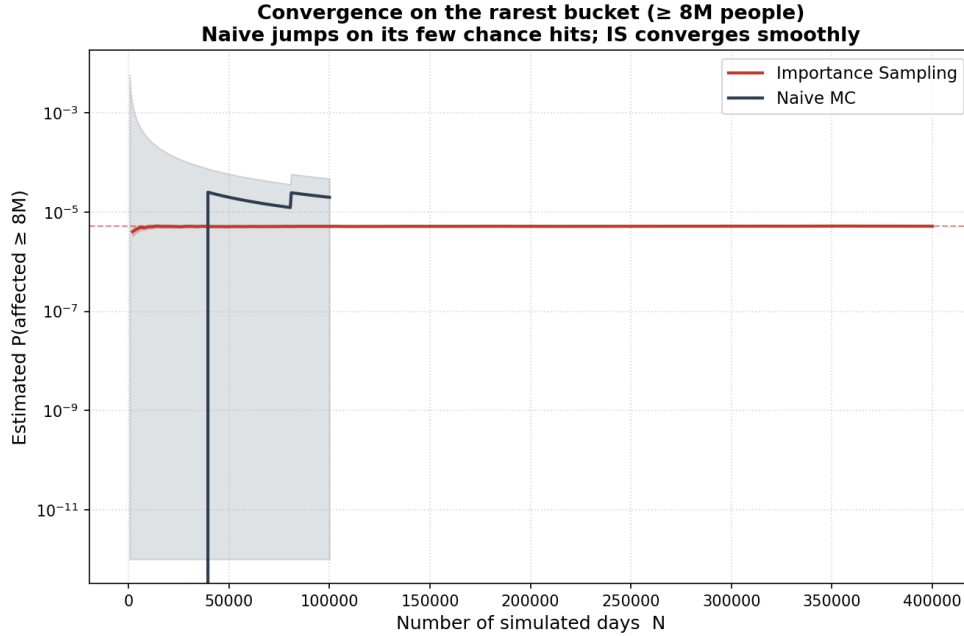
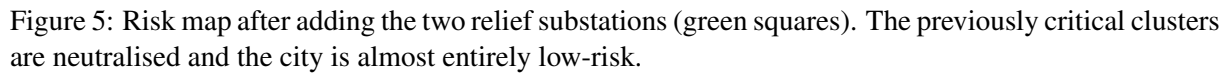


Figure 3: Convergence of the estimate for $P(\text{affected} \geq 8\,000\,000)$. Importance Sampling converges smoothly; the naive estimate only jumps on its few chance hits and its error band still spans zero.

5.4 Where the Risk Lies

Combining the per-district failure probabilities with the populations gives the expected number of people without power on an average day, which is about 68 000. Figure 4 maps the per-district blackout probability. The risk is strongly concentrated: three low-margin districts dominate, Maltepe, Beyoğlu, and Fatih, each failing on about 5.2% of days – exactly the districts that the EPIAŞ data flagged as having the most overloads, and all carrying the minimum safety margin $\alpha = 0.08$. Every other district sits at least an order of magnitude lower, and most of the city is green (negligible risk). This concentration is what makes targeted reinforcement, studied next, so effective.

Metric	Baseline	With 2 stations	Improvement
Expected daily affected (people)	67664	4136	−93.9%
$P(\text{affected} \geq 1\,000\,000)$	2.39×10^{-2}	5.60×10^{-4}	$\sim 43\times$
$P(\text{affected} \geq 8\,000\,000)$	5.22×10^{-6}	8.77×10^{-7}	$\sim 6\times$



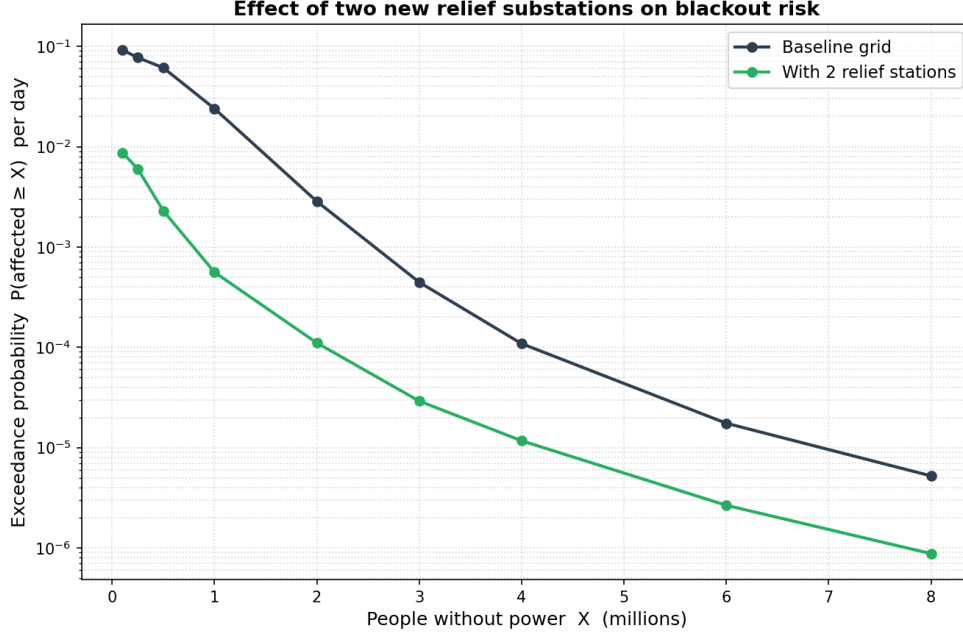


Figure 6: Exceedance curves before and after adding the two relief stations. The curve drops by orders of magnitude across all event sizes.

7 Limitations and Assumptions

The model is intentionally simplified, and its absolute probabilities should be read as illustrative rather than predictive. The main assumptions are:

- No power-flow physics. The grid is a graph with a load-shedding rule, not a true electrical model: there are no line impedances, no Kirchhoff laws, and no voltage or frequency dynamics. The adjacency is *geographic* (which districts border each other), which is only a proxy for the real electrical transmission topology.
- Demand model. Daily demand is Gaussian about a population-scaled mean. Real demand has daily and seasonal cycles and heavier tails than a normal distribution.
- Calibration. District loads are split by population, and safety margins are mapped from overload counts by a linear heuristic; neither uses confidential utility capacity data.
- Cascade rule. Redistribution is a simple headroom-weighted split with a sharp breaker threshold. Real cascades involve protection relays, controlled islanding, and generator dynamics.

Because of these choices, the value of the work lies in the *methodology* (rare-event estimation by importance sampling) and in the *relative* conclusions (which districts dominate the risk, and where reinforcement helps most), rather than in the precise numerical probabilities. Most of these limitations are also natural directions for making the model more realistic.

8 Conclusion

This project modelled cascading blackouts in the Istanbul mainland grid as a Monte Carlo problem, with district safety margins calibrated from real EPIAŞ overload data. The central finding is methodological: a city-wide blackout is a genuinely rare event, and naive Monte Carlo cannot resolve it – in 100000 simulated days it logged only two events affecting more than 8 million people, an estimate so dominated

by sampling noise that its 95% interval runs from near zero to several times the true value. By biasing a single systemic demand variable and reweighting, Importance Sampling estimated the same probability as $(5.22 \pm 0.07) \times 10^{-6}$, with a variance reduction of up to about 10000 $\times$, while agreeing with the naive method on the common events that validate it. Finally, the risk map showed that vulnerability is concentrated in a few historically weak districts, and a greedy placement of two relief substations on those clusters cut the expected affected population by about 94%. The work demonstrates both why variance reduction is necessary for rare-event risk and how the resulting estimates can directly inform an engineering decision.

References

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